

Project 4

Lottery System with Commit-Reveal Randomness

Field	Details
Course	CS218 — Programmable and Interoperable Blockchain
Project	Project 4 — Lottery System with Commit Reveal Randomness
Toolchain	Foundry · Solidity · OpenZeppelin

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1. Problem Statement & Blockchain Justification

1.1 The Problem

Traditional lotteries suffer from a fundamental trust problem: the organiser controls the random draw, creating an opportunity for manipulation. Participants must trust a central authority to select the winner fairly, with no way to verify the outcome independently. So we need to trust payment processors, custodians, and prize administrators — each a separate point of failure.

Our project solves this using a decentralised, commit-reveal lottery deployed on Ethereum. All participants, funds, and randomness selection logic are governed entirely by a transparent smart contract. No single party — not even the contract owner — can manipulate the outcome once the commit is on-chain.

1.2 Why Blockchain Is the Right Tool

Blockchain is uniquely suited to this problem for four reasons:

- Trustless execution: Smart contract code runs identically for all participants. No intermediary can alter or delay the draw.

- Transparent fund management: All ETH deposits, the prize pool, and withdrawals are publicly verifiable on-chain. Participants can audit the contract balance at any time.
- Commit-reveal randomness: Ethereum does not have a native secure random oracle. The commit-reveal scheme — submitting a hash of a secret before revealing it — is the canonical on-chain solution. It prevents the owner from changing the secret after seeing the pool, while the future blockhash prevents the owner from predicting or biasing the winner index before the reveal window.
- Censorship resistance: Ticket purchase, prize claim, and refund functions are publicly callable. No one can block a winner from collecting their prize.

A traditional database or web server cannot provide these guarantees. An off-chain randomness oracle (e.g., Chainlink VRF) would be an alternative, but the commit-reveal scheme demonstrates the underlying cryptographic principle without external dependencies.

2. Architecture & Design

2.1 Phase Table

The contract enforces a strict, linear phase progression per round. Each function is gated to only execute in the correct phase:

Phase	Entry Trigger	Allowed Actions	Next Phase
Open	Constructor / startNewRound()	buyTicket()	SaleClosed
SaleClosed	closeSale()	commitHash()	Committed
Committed	commitHash()	revealAndDraw() or slashOwner()	Drawn or Slashed
Drawn	revealAndDraw()	claimPrize()	(next round starts)
Slashed	slashOwner()	claimRefund()	(next round starts)

The linear phase flow is: Open → SaleClosed → Committed → Drawn (normal path) or Committed → Slashed.

2.2 Multi-Round Architecture

The contract supports unlimited consecutive rounds. Each round is fully isolated via the currentRound counter. All round-specific state (participants, prize pool, committed hash, winner, collateral) is stored in a mapping keyed by round ID, so historical rounds remain readable and claimable without interference from new rounds.

Key design decision: startNewRound() requires the previous round to be in Drawn or Slashed phase. This prevents the owner from abandoning an active round and avoids prize pool corruption across rounds.

2.3 Commit-Reveal Randomness

The commit-reveal scheme works in two steps:

1. Commit: The owner generates a random secret off-chain and submits `keccak256(abi.encodePacked(secret))` via `commitHash()`. This locks the owner into a specific

outcome without revealing it to the mempool. To prevent griefing, the owner must also lock 1 ETH as collateral. This transaction establishes a targetBlock (current block + 10).

2. **Reveal:** After the targetBlock is mined, the owner calls revealAndDraw(secret). The contract verifies the secret against the committed hash. The winner is then calculated using:

```
uint256 winnerIndex = uint256(keccak256(abi.encodePacked(secret,
blockhash(targetBlock), totalParticipants))) % totalParticipants;.
```

Why NOT block.timestamp alone: A validator who has purchased a ticket can choose which timestamp to use when proposing the draw block, biasing the result toward their own address. The commit-reveal scheme eliminates this: by the time the secret is revealed, the winning blockhash is already finalised and immutable. The owner is committed to a specific secret before the blockhash is known, and the blockhash is finalised before the secret is disclosed — neither party alone controls the outcome.

Residual trust: While this prevents validators from manipulating the draw, it leaves one residual trust vector: the owner knows the secret. If the owner calculates the winner off-chain and dislikes the result, they could simply refuse to call revealAndDraw().

We neutralize this via a financial penalty. If the owner fails to reveal within a 250-block window, the slashOwner() function unlocks. This permanently burns/redistributes the owner's 1 ETH collateral, allowing participants to call claimRefund() to recoup their ticket price plus a proportional cut of the owner's penalty.

2.4 Contract Functions Overview

Function	Caller	Phase Required	Description
constructor(ticketPrice)	Deployer	—	Sets ticket price; opens Round 1
buyTicket()	Anyone	Open	Registers msg.sender as participant; adds ETH to prize pool
closeSale()	Owner	Open	Ends ticket sales; requires ≥1 participant
commitHash(hash)	Owner	SaleClosed	Submits keccak256(secret); sets targetBlock; locks collateral
revealAndDraw(secret)	Owner	Committed	Verifies hash; selects winner; returns collateral to owner
slashOwner()	Anyone	Committed	Callable after 250 blocks without reveal; triggers refund phase
claimPrize(roundId)	Winner	Drawn	Winner withdraws full prize pool via CEI pattern
claimRefund(roundId)	Participant	Slashed	Proportional refund of ticket cost + share of collateral
startNewRound()	Owner	Drawn/Slashed	Increments round ID and resets to Open phase
totalTickets(roundId)	Anyone	Any	View: returns participant count for a round

getParticipant(roundId, idx)	Anyone	Any	View: returns participant address at given index
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3. Smart Contract — Functions & Correctness

3.1 Constructor

The constructor sets up the initial state by taking a single argument for `_ticketPrice`. It validates that the price is greater than zero to prevent free entries, initializes `currentRound` to 1, and opens the first phase. We inherit OpenZeppelin's `Ownable` here, immediately setting `msg.sender` as the contract owner to establish our baseline access control.

3.2 buyTicket()

This is the primary entry point for users. To prevent accounting mismatches and token drain vulnerabilities, we handle two critical edge cases immediately:

1. The transaction reverts if `msg.value` does not *exactly* equal `ticketPrice` (no partial payments or overpayments accepted).
2. The transaction reverts if the current phase is not strictly Open (preventing late entries after the pool is locked).

On success, `msg.sender` is appended to the `participants` array. We allow the same address to buy multiple tickets—each purchase pushes another entry into the array, naturally increasing their probability of winning the modulo calculation during the draw. We also increment `userTickets` to track individual balances in case a proportional refund is triggered later.

3.3 closeSale()

Access Control: Restricted via `onlyOwner`

Requires at least one participant exists (to prevent a meaningless round). Transitions phase to `SaleClosed`. Emits `SaleClosed`.

3.4 commitHash(bytes32 hash)

Access Control: Restricted via `onlyOwner`. Must be called after `closeSale()`.

This function forces the owner to lock collateral exactly equal to the current prize pool (`require(msg.value == round.prizePool)`). This ensures that the owner has as much at stake as all participants combined, creating a symmetric financial incentive to reveal honestly.

Edge Cases Handled: We explicitly require that the provided hash is not `bytes32(0)` to prevent a null-commit exploit.

Why 10 blocks: This gives the owner a predictable but short window to wait for the target block to be mined before revealing. The 250-block ceiling ensures `blockhash()` remains available (EVM only stores the last 256 block hashes).

3.5 revealAndDraw(bytes32 secret)

Access Control: Restricted via `onlyOwner`.

Three block-number checks guard this function:

- `block.number` must be $>$ `targetBlock` (the target block is mined)
- `block.number` must be $\leq$ `targetBlock + 250` (`blockhash` still available)
- `keccak256(abi.encodePacked(secret))` must equal `committedHash` (integrity check)

On success, the winner index is computed deterministically:

```
uint256 winnerIndex =  
uint256(keccak256(abi.encodePacked(_secret, blockhash(targetBlock[currentRound]),  
    totalParticipants)) ) % totalParticipants;
```

The winner address is stored, phase transitions to Drawn, the locked collateral is returned to the owner (reward for honest reveal), and WinnerDrawn is emitted. The checks-effects-interactions pattern is followed: state changes occur before the collateral ETH transfer.

3.6 slashOwner()

Callable by anyone after `block.number > targetBlock + 250` in the Committed phase. Sets phase to Slashed. No ETH is transferred here — participants call `claimRefund()` individually to pull their share. Emits OwnerSlashed.

3.7 claimPrize(uint256 roundId)

Protected by nonReentrant. Validates: phase is Drawn, `msg.sender` equals `winner[roundId]`, and prize has not been claimed yet. Follows CEI: `prizePool` zeroed and `prizeClaimed` set to true before the ETH call. Emits PrizeClaimed.

3.8 claimRefund(uint256 roundId)

Protected by nonReentrant. Valid only in Slashed phase. Calculates the caller's proportional share:

```
uint256 totalFundsAvailable = prizePool[roundId] + lockedCollateral[roundId];  
uint256 refundAmount = (tickets * totalFundsAvailable)  
    / participants[roundId].length;
```

This ensures every participant receives their ticket cost back plus a bonus share of the owner's slashed collateral. `userTickets` is zeroed before the transfer to prevent double-claiming.

3.9 Access Control

The following access control rules are enforced:

Function	Restriction	Enforcement
closeSale, commitHash, revealAndDraw, startNewRound	Owner only	onlyOwner modifier (Ownable)
claimPrize	Winner of that round only	require(msg.sender == winner[roundId])
claimRefund	Ticket holder in Slashed round	require(userTickets[roundId][msg.sender] > 0)
slashOwner, buyTicket	Anyone (permissionless)	No caller restriction; phase gated

4. Security Analysis

4.1 Reentrancy Guard

The two ETH-sending functions that could be vulnerable to reentrancy are protected:

Function	ETH Transfer	Protection
claimPrize()	Sends full prize pool to winner	nonReentrant modifier + CEI pattern
claimRefund()	Sends proportional refund to participant	nonReentrant modifier + CEI pattern
revealAndDraw()	Returns collateral to owner	CEI pattern (owner is trusted; nonReentrant not applied here)

The checks-effects-interactions (CEI) pattern is followed throughout: all state variables are updated before any external ETH call is made. In `claimPrize()`, `prizePool` is zeroed and `prizeClaimed` is set to true before the `call{value: payout}()` is executed. Even if a malicious winner contract attempted a re-entry, the `nonReentrant` modifier's lock flag would cause the re-entry call to revert immediately.

4.2 Input Validation

All public state-changing functions include comprehensive `require()` statements. Key validations:

- `msg.value == ticketPrice` — exact match enforced; no rounding or tolerance
- `_hash != bytes32(0)` — prevents a trivially guessable empty commit
- `msg.value > 0` in `commitHash` — ensures owner has skin in the game
- `participants.length > 0` before `closeSale` and `commitHash` — prevents empty-round exploits
- Phase checks on every function — incorrect sequencing is always rejected

No overflow or underflow is possible: the contract uses Solidity ^{^0.8.20}, which has built-in checked arithmetic by default.

4.3 Visibility Modifiers

Visibility has been carefully assigned across all functions and state variables:

Identifier	Visibility	Reason
participants (array per round)	private	Accessed via <code>getParticipant()</code> and <code>totalTickets()</code> view helpers
rounds (in <code>gasOptimisedLottery</code>)	private	Exposed via individual view functions; prevents arbitrary struct access
<code>ticketPrice</code> , <code>currentRound</code>	public	Read by frontend and participants to verify contract state
<code>buyTicket</code> , <code>claimPrize</code> , <code>claimRefund</code>	external	Only callable from outside; marginally cheaper than public
<code>totalTickets</code> , <code>getParticipant</code>	external view	Read-only; no state mutation

4.4 Front-Running Resistance

The commit-reveal scheme directly addresses the front-running / miner bias problem:

- Before commit: The secret is unknown to everyone on-chain. No one can compute the future winner.
- After commit, before reveal: The hash is public, but the preimage (secret) is unknown. The winner cannot be determined.

- At reveal: The secret and the target blockhash are both fixed. The winner is computed deterministically — no party can influence either value at this point.

The 250-block window creates a time-bounded incentive to reveal promptly. The collateral deposit ensures the reveal is economically rational for an honest owner.

5. OpenZeppelin Usage

5.1 Contracts Used

OZ Contract	Applied To	Benefit
Ownable	Lottery contract	Provides onlyOwner modifier; manages owner address; emits OwnershipTransferred event; avoids hand-rolled access control bugs
ReentrancyGuard	claimPrize(), claimRefund()	Provides nonReentrant modifier with a battle-tested reentrancy lock; prevents re-entrant ETH withdrawal exploits

Both contracts are imported from @openzeppelin/contracts v5.x, the current audited release. The import paths used in the source:

```
import {Ownable} from "@openzeppelin/contracts/access/Ownable.sol";
import {ReentrancyGuard} from "@openzeppelin/contracts/utils/ReentrancyGuard.sol";
```

Why not write our own: Hand-rolled Ownable and ReentrancyGuard implementations introduce well-known vulnerabilities (e.g., storage slot collisions, incorrect modifier ordering, or missing initialisation in constructors). The OpenZeppelin versions have been audited by multiple independent security firms and are used in protocols managing billions of dollars of value.

6. Gas Optimisation

6.1 Gas Reports

Gas reports were generated using the Foundry toolchain:

```
forge test --gas-report
```

The table below summarises the key functions before and after optimisation:

BEFORE — lottery.sol (unoptimised)

Function Name	Min	Avg	Median	Max	# Calls
buyTicket	26,006	90,968	84,865	119,065	1,628
claimPrize	29,666	33,757	30,051	39,606	5
claimRefund	29,644	35,970	31,869	43,234	5
closeSale	23,836	49,534	51,663	51,663	34

commitHash	24,537	92,045	102,236	102,236	29
revealAndDraw	24,517	44,121	53,788	53,788	17
slashOwner	25,967	30,891	32,395	32,395	7
startNewRound	23,815	31,093	31,844	35,371	6

AFTER — gasOptimisedLottery.sol (optimised)

Function Name	Min	Avg	Median	Max	# Calls
buyTicket	26,006	91,113	85,865	119,065	1,590
claimPrize	29,666	33,757	30,051	39,606	5
claimRefund	29,644	35,970	31,869	43,234	5
closeSale	23,836	49,534	51,663	51,663	34
commitHash	24,537	92,045	102,236	102,236	29
revealAndDraw	24,517	44,031	53,635	53,635	17
slashOwner	25,967	30,891	32,395	32,395	7
startNewRound	23,815	31,093	31,844	35,371	6

6.2 Identifying the Most Expensive Function

Ranked by average gas cost (before optimisation):

Rank	Function	Avg Gas	Call Count
1st	commitHash	92,045	29 (one per round — owner only)
2nd	buyTicket	90,968	1,628 (every participant)
3rd	revealAndDraw	44,121	17 (one per round — owner only)
4th	claimPrize	33,757	5 (once per round — winner only)

While commitHash has the highest single-call average (92,045 gas), it is a fixed one-time cost per round paid by the owner. The function specifically targeted for optimisation is revealAndDraw. It is the most computationally heavy function in the lottery lifecycle, containing complex keccak256 hashing, blockhash lookups, and multiple state reads — making it the ideal candidate for targeted EVM-level optimisation.

6.3 Optimisation Applied — gasOptimisedLottery.sol

Both lottery.sol and gasOptimisedLottery.sol share the same Round struct architecture. The optimisation applied in gasOptimisedLottery.sol is targeted specifically at the revealAndDraw() function using two EVM-level techniques

Metric	Before (lottery.sol)	After (gasOptimiseLottery.sol)	Gas Saved
Min Gas	24,517	24,517	0
Avg Gas	44,121	44,031	90
Median Gas	53,788	53,635	153
Max Gas	53,788	53,635	153

Why revealAndDraw Was Targeted

In the EVM, interacting with state variables stored on the blockchain (SLOAD operations) is expensive:

- A "cold" read (first access in a transaction) costs 2,100 gas.
- A "warm" read (subsequent access in the same transaction) costs 100 gas.
- Reading a variable from local memory costs roughly ~3 gas.

Additionally, Solidity 0.8.0+ introduces built-in safety checks for math operations, including a zero-divisor check (ISZERO, PUSH, JUMPI opcodes) before every division or modulo operation, adding unnecessary overhead when the divisor is already known to be non-zero.

A. Removing Redundant Math Checks (unchecked block)

In the original contract, the winner index modulo (% totalParticipants) triggers a Solidity-injected zero-divisor safety check. However, the contract's phase architecture guarantees totalParticipants > 0 because closeSale() explicitly requires it before the lottery can ever reach Committed or Drawn phases. By wrapping the modulo operation in an unchecked { } block, we explicitly instruct the compiler to drop the zero-divisor check, saving the overhead of those opcodes:

```
uint256 winnerIndex; unchecked{winnerIndex=uint256(keccak256(abi.encodePacked(_secret,
future BlockHash, totalParticipants))) % totalParticipants; }
```

B. Explicit Memory Caching (Preventing Stack Shuffling Penalties)

When complex block scopes (like the unchecked block) are introduced, the Solidity compiler's optimizer can lose track of storage pointers, forcing expensive "warm" SLOAD re-reads and triggering stack-shuffling operations (SWAP, DUP). To prevent this, state variables are read from storage exactly once and cached in local memory:

- Caching targetBlock as _targetBlock: Used in require statements and to fetch the blockhash.
- Caching winner as _winner: Passed into WinnerDrawn event without re-reading from storage.
- Caching prizePool as _prizePool: Read prior to the external call and event emission.

Total savings: 153 gas reduction in maximum execution cost and 90 gas reduction in average cost per call.

7. Testing

7.1 Test Suite Summary

The test suite is written in Solidity using Foundry's forge-std framework (test/lotterytest.t.sol). All 52 tests pass with zero failures.

```
forge test --match-path test/lotterytest.t.sol
```

```
Ran 52 tests for test/lotterytest.t.sol:LotteryTest
Suite result: ok. 52 passed; 0 failed; 0 skipped
```

7.2 Coverage Report

Coverage was measured using:

```
forge coverage --match-path src/lottery.sol
```

File	% Lines	% Statements	% Branches	% Functions
lottery.sol	100% (80/80)	100% (76/76)	90.38% (47/52)	100% (11/11)
Total	100%	100%	90.38%	100%

Line and function coverage are both 100%. Branch coverage is 90.38% (47/52) — the uncovered branches correspond to implicit else paths in Solidity's `require()` checks (where the condition is always true in the successful test path). These branches are exercised by the corresponding revert tests but are not reachable as a 'false' branch in the success path, which is expected behaviour.

8. Chainlink VRF Integration

To demonstrate an alternative approach to decentralized randomness, we developed a parallel implementation of the lottery (`LotteryVRFChainlink.sol`) that utilizes the Chainlink Verifiable Random Function (VRF) v2.5 oracle network.

8.1 Eliminating Residual Trust (Why VRF?)

While our primary commit-reveal architecture successfully prevents miner front-running, it leaves a "residual trust" vector: the owner knows the secret and could technically refuse to reveal it if they dislike the outcome. We mitigated this in the base contract via a financial slashing penalty.

Chainlink VRF eliminates this trust vector entirely. The randomness is generated off-chain by an oracle node and verified on-chain via cryptographic proofs before it is consumed by the contract. Because the contract owner is entirely removed from the randomness generation process, we completely eliminated the need for owner collateral deposits.

8.2 Architectural Overhaul (How it Works)

Integrating VRF required restructuring the contract's inheritance and lifecycle:

- **Ownership:** We removed OpenZeppelin's `Ownable` because Chainlink's `VRFConsumerBaseV2Plus` natively implements its own `ConfirmedOwner` pattern. Inheriting both creates function clashes.

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- The Request Phase: The `commitHash()` function is replaced by `requestDraw()`. The owner calls this function to trigger a request to the `s_vrfCoordinator`, asking for 1 random word with a 3-block confirmation requirement and a 100,000 gas callback limit.
 - The Callback Phase: The `revealAndDraw()` function is replaced by the internal `fulfillRandomWords()` callback. Chainlink nodes call this automatically once the randomness is cryptographically verified.

Once the VRF coordinator verifies the proof, it deterministically selects the winner without human intervention:

8.3 Slashing as an Oracle Fallback

In the commit-reveal model, `slashOwner()` punished a malicious owner. In the VRF model, it serves as a network safety fallback. If the Chainlink network suffers an outage or the VRF request fails to fulfill within 256 blocks, the contract could theoretically freeze in the Committed phase. The adapted `slashOwner()` function acts as a timeout, allowing anyone to push the contract into the Slashed phase so users can reclaim their ticket funds.

8.4 VRF Testing and Coverage

Testing oracle callbacks requires specialized mock contracts. We utilized Foundry's `VRFCoordinatorV2_5Mock` to simulate the oracle environment locally. By using `vrfCoordinator.fulfillRandomWords()` within our test suite, we successfully simulated the async callback process.

The VRF implementation passed all 45 integration tests and achieved perfect execution coverage:

- Line Coverage: 100.00% (72/72)
- Statement Coverage: 100.00% (66/66)
- Branches Coverage: 89.47% (34/38)
- Function Coverage: 100.00% (11/11)

9. DApp Frontend

9.1 Overview

The frontend is a React application that connects to the deployed Lottery contract via `ethers.js` / `wagmi`. It supports MetaMask wallet connection, live contract state display, and ticket purchase.

9.2 Wallet Connection

On load, the application checks for `window.ethereum` (MetaMask provider). The user clicks Connect Wallet, which triggers an `eth_requestAccounts` call. On success, the connected address and ETH balance are displayed in the header. The contract ABI is bundled from the Foundry build output (`out/lottery.sol/Lottery.json`).

9.3 Contract State Display

The UI reads and displays the following contract state on every block (via `ethers.js` `provider.on('block', ...)`):

- Current round number
- Current phase (Open / SaleClosed / Committed / Drawn / Slashed)

- Prize pool (in ETH)
- Total tickets sold
- Ticket price
- Winner address (if phase is Drawn)

9.4 State-Changing Function — buyTicket()

To prevent users from wasting gas on transactions that will inevitably revert, the frontend conditionally disables the "Buy Ticket" button unless the contract phase is strictly Open and the connected wallet holds sufficient ETH.

The transaction lifecycle is handled as follows:

1. The React app calls `contract.buyTicket({ value: ticketPrice })` via the `ethers.js` instance.
2. MetaMask intercepts the call, prompting the user to confirm the transaction and sign the exact ETH payload.
3. The UI renders a loading spinner while waiting for the transaction to be mined into a block.
4. Upon confirmation (`tx.wait()`), a success toast displays the transaction hash, and the frontend automatically polls the new prize pool and ticket count.
5. If the transaction fails (e.g., user rejected, or phase shifted to Closed), the specific revert reason from the Solidity contract is parsed and displayed to the user.

10. On-Chain vs Off-Chain Data

10.1 Data Placement Decisions

Stored ON-CHAIN (required for contract logic)	Kept OFF-CHAIN (never stored on-chain)
Participant wallet addresses (required for winner selection and refunds)	The raw secret value — never written to storage. Passes through calldata transiently only at reveal time, then discarded.
Phase enum per round (enforces state machine transitions)	Any personal details, KYC documents, or names linked to wallet addresses
Committed hash — <code>bytes32 keccak256(secret)</code> — the cryptographic commitment	Lottery promotional materials, ticket artwork, event descriptions (store on IPFS; put CID on-chain if needed)
Prize pool amount in wei (must be auditable by all participants)	Owner's off-chain secret — generated locally before <code>commitHash()</code> is called
Winner address per round (needed for <code>claimPrize()</code> authorisation)	User preferences, UI settings, or any data not required for contract logic
Target block number per round (needed to validate reveal timing)	
Locked collateral per round (needed for slash-and-refund calculation)	

10.2 GDPR Compliance

Blockchain state is strictly append-only. Because it is technologically impossible to delete data once it is mined, storing any Personal Identifiable Information (PII) — like names, emails, or IDs — directly on-chain permanently violates a user's right to erasure under GDPR Article 17.

To ensure strict compliance, we designed this lottery to be completely pseudonymous. The contract only ever interacts with raw 0x wallet addresses. Any external data, such as UI preferences or potential KYC verifications, is handled entirely off-chain.

10.3 Gas Cost Implication

The commit-reveal pattern is not just a security necessity; it is a massive gas optimization. Writing a new 32-byte storage slot (SSTORE) costs a flat 20,000 gas.

If we stored the raw secret permanently on-chain, it would incur this maximum penalty. Instead, the raw secret is only ever passed transiently as calldata during the `revealAndDraw()` execution. Because the EVM only charges 16 gas per non-zero calldata byte, passing a 32-byte secret costs approximately 512 gas. This approach is roughly **39x cheaper** than a state write, while simultaneously eliminating the risk of mempool front-running.

11. Conclusion

The development of this project successfully demonstrates how to architect secure, trustless, and economically viable decentralized lotteries on Ethereum. By engineering two distinct solutions — a native cryptographic commit-reveal protocol and a Chainlink VRF oracle integration — we proved multiple pathways to eliminate miner front-running vulnerabilities and guarantee fair randomness.

Beyond functional correctness, this project prioritized rigorous engineering standards. Through targeted struct packing and storage optimization, we reduced the execution cost of the highly computational `revealAndDraw()` function. By caching state variables like `targetBlock` and `prizePool` into local memory, we prevented redundant 100-gas SLOAD operations, proving that security does not have to come at the expense of EVM efficiency.

Both contract variations are fortified against DeFi's most common attack vectors through strict adherence to the Checks-Effects-Interactions pattern, pull-over-push refund mechanics, and OpenZeppelin's ReentrancyGuard. Our Foundry test suites verify this resilience with 100% statement and line coverage across the board.

Finally, the system's architecture addresses the fundamental friction between immutable public ledgers and real-world legal frameworks. By strictly relegating non-essential data and personal identifiers to off-chain environments, the protocol remains strictly pseudonymous. This ensures absolute compliance with GDPR Article 17 while simultaneously avoiding the prohibitive 20,000 gas SSTORE penalties. Ultimately, this project serves as a production-ready blueprint for handling high-stakes, time-sensitive, and mathematically provable logic on a public blockchain.